MTH203: Sequences and series

Homework 06

29/11/2025, 6:00 PM

(Note: This homework will only be graded for "completion". It is worth 3 times each of the earlier homeworks. So, it is worth 3.75 % of your total course performance. Solutions will *not* be uploaded, but you are welcome to ask for help. Late submissions will NOT be accepted.)

(1) Use the formal "$\epsilon - N$" definition of a limit to prove the following statements:

  (a) Using the formal $\epsilon - N$ definition of a limit, prove that

$$\lim_{n\to\infty} \frac{5n+3}{n-8} = 5$$

  (b) Using the formal $\epsilon - N$ definition of a limit, prove that

$$\lim_{n\to\infty} \frac{1}{\sqrt{n+1}} = 0$$

  (In this problem, you must not use the theorems regarding algebraic properties of limits. The purpose of this problem is to test your understanding of the definition of limits.)

(2) Let x be a positive real number such that $x \geqslant 1$. Prove that $\lim_{n\to\infty} x^{(1/n)} = 1$. (What happens if $0 < x < 1$?)

(3) Calculate the following limits (use any method that is taught in the course):

  (a) $\lim_{n\to\infty} \frac{\sqrt{9n^2+n}}{2n-1}$

  (b) $\lim_{n\to\infty} \sqrt{n^2+4n} - n$

  (c) $\lim_{n\to\infty} \frac{n^2\cos(n!)}{n^4+1}$

  (d) $\lim_{n\to\infty} \left(\frac{n+1}{n}\right)^{1/n}$

(4) If $(a_n)_{n\in\mathbb{N}}$ is a bounded sequence and $(b_n)_{n\in\mathbb{N}}$ converges to 0, prove that $(a_nb_n)_{n\in\mathbb{N}}$ converges to 0.

(5) Prove that if a sequence $(a_n)_{n\in\mathbb{N}}$ converges to a limit L, then the sequence of $(c_n)_{n\in\mathbb{N}}$ defined by $c_n = \frac{a_1+a_2+\cdots+a_n}{n}$ also converges to L. (This is a slightly harder problem than the rest.)

(6) Let (x_n) be a sequence satisfying the property that $|x_{n+1} - x_n| \leqslant \frac{1}{3^n}$ for all $n \in \mathbb{N}$. Prove that (x_n) is a Cauchy sequence.

(7) Prove that the following sequences converge and calculate the limits:

  (a) The sequence $(a_n)_{n\in\mathbb{N}}$, where $a_1 = 5$ and $a_{n+1} = \frac{1}{2}\left(a_n + \frac{9}{a_n}\right)$ for $n \geqslant 1$.

  (b) The sequence $(b_n)_{n\in\mathbb{N}}$ where $b_1 = 0$ and $b_{n+1} = \sqrt{10 + c_n}$ for $n \geqslant 1$.

(8) Consider the sequence $(x_n)_{n\in\mathbb{N}}$, where $x_1 = 1$ and $x_{n+1} = 1 + \frac{1}{x_n}$ for $n \geqslant 1$.

  (a) Use induction to show that $1 \leqslant x_n \leqslant 2$ for all $n \in \mathbb{N}$. Also prove that $3/2 \leqslant x_n \leqslant 2$ for $n > 1$.

  (b) Prove that $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence.

(9) Prove: If the subsequence $(x_{2k-1})_{k\in\mathbb{N}}$ (odd terms) and the subsequence $(x_{2k})_{k\in\mathbb{N}}$ (even terms) of a sequence $(x_n)_{n\in\mathbb{N}}$ both converge to the **same** limit L, then $(x_n)_{n\in\mathbb{N}}$ also converges to L.

(10) Prove that the sequence $a_n = \sqrt{n+1} - \sqrt{n}$ is monotone and bounded. Calculate the limit of this sequence.

(11) For what values of x does the series $\sum_{n=0}^{\infty} 2^n (x+1)^n$ converge? Find the sum of the series for these values of x.

(12) Determine if the following series converge or diverge:

(a) $\sum_{n=1}^{\infty} \frac{n^2}{n^4+3}$

(b) $\sum_{n=1}^{\infty} \frac{1}{n3^n}$

(c) $\sum_{n=1}^{\infty} \frac{3n+5}{\sqrt{n^5+2n}}$

(d) $\sum_{n=2}^{\infty} \frac{(-1)^n n}{n^2-1}$

(13) Use the Ratio Test to determine if $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges or diverges.

(14) Use the Root Test to determine if $\sum_{n=1}^{\infty} \left(\frac{n+1}{3n-1}\right)^n$ converges or diverges.

(15) Classify the following series as absolutely convergent, conditionally convergent or divergent.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$

(b) $\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^4+2}$

(16) Prove or disprove: If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} \frac{a_n}{n}$ also converges absolutely.

(17) Prove or disprove: If $\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} b_n$ converges, then the series formed by the product of terms, $\sum_{n=1}^{\infty} a_n b_n$, must converge.

Hints:

1. (a) Divide both the numerator and denominator by n.
   (b) Remember, you are supposed to compute the limit from "scratch". So, give $\epsilon > 0$, how large does n need to be so that $\frac{1}{\sqrt{n+1}} < \epsilon$?

2. Suppose Prove that the sequence $(x^{1/n})_{n\in\mathbb{N}}$ is monotone and that it is bounded below by 1. Thus, it has a limit, which we denote by α. Then $\alpha < x^{1/n}$ for all $n \in \mathbb{N}$. So, $\alpha^n < x$ for all n. Deduce that $\alpha = 1$.

3. (a) Divide both the numerator and denominator by n.
   (b) Multiply and divide by the "conjugate".
   (c) Try to use the Squeeze Theorem. What is the maximum and minimum value that $\cos(x)$ can take?
   (d) Use the Squeeze Theorem. Use $1 \leqslant (1 + 1/n)^{1/n} \leqslant 2^{1/n}$. Use Problem 2.

4. Should be an easy "$\epsilon - N$" argument.

5. Let $\epsilon > 0$ be given and let N be such that if $n > N$, $|c_n - L| < \epsilon$.
   Write $c_n - L$ as $\sum_{k=1}^{n} \frac{c_k - L}{n}$ and use the triangle inequality.
   You will have to estimate $\sum_{k=1}^{n} \frac{|c_k - L|}{n}$.
   You will need to split this sum as follows:
   $$\sum_{k=1}^{n} \frac{|c_k - L|}{n} = (1/n)\left[\left(\sum_{k=1}^{N} |c_k - L|\right) + \left(\sum_{k=N+1}^{n} |c_k - L|\right)\right]$$
   Every term in the second sum is smaller than ϵ. Now, see what happens when n becomes very large. The first sum is does not depend on n, so it is fixed. It is only the second sum that is changing as n becomes large.

6. In both parts of the problem, you can establish that the sequences are monotone and it is easy to show that they are bounded. Now use the Monotone Convergence Theorem to conclude that they converge.

7. You will need to use the fact that $\sum_{k=1}^{\infty} 1/3^k$ is absolutely convergent.

8. Part (a) is very easy. For part (b), use the fact that for any $n \geqslant 1$, we have
   $$|x_{n+1} - x_{n+2}| = \frac{|x_n - x_{n+1}|}{|x_n x_{n+1}|}.$$
   Since $n \geqslant 1$, by part (a), $|x_n| \geqslant 1$ and $|x_{n+1}| \geqslant 1$. Thus, we have $|x_{n+1} - x_{n+1}| \leqslant \frac{|x_n - x_{n+1}|}{3/2}$. Conclude that $(|x_{n+1} - x_n|)_{n\in\mathbb{N}}$ is a strictly decreasing sequence that converges to 0. Conclude that $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence as in Problem 7.

9. This is an easy $\epsilon - N$ argument.

10. Multiply and divide by the "conjugate".

11. Easy application of formula for convergence of power series.

12. In all these problems, divide the numerator and denominator by a suitable power of n. In part (d), you need to use the Alternating Series Test.

(No hints for the remaining problems. They are very easy.)